

Transformer Model Q&A

1. Q: What is a token in language models?

A: A token is a unit of text, such as a word or subword, used as input to language models. Tokenization splits text into tokens.

2. Q: What is embedding in the context of Transformers?

A: Embedding is a vector representation of tokens that captures their semantic meaning in a continuous space.

3. Q: What is self-attention?

A: Self-attention is a mechanism that allows each token in a sequence to attend to every other token, enabling contextual understanding.

4. Q: What is the purpose of positional encoding?

A: Positional encoding injects information about the position of tokens in a sequence since the Transformer has no recurrence.

5. Q: Why use Transformers over RNNs?

A: Transformers enable parallelization, handle long-range dependencies better, and scale more efficiently than RNNs.

6. Q: What is multi-head attention?

A: Multi-head attention allows the model to focus on different positions and aspects of the input simultaneously.

7. Q: What is the role of the feed-forward network in Transformer blocks?

A: It processes the output of the attention mechanism and helps capture complex transformations of the data.

8. Q: How is the attention score computed?

A: It is computed using the dot product of the query and key vectors, scaled, and passed through a

softmax to get attention weights.

9. Q: What are Q, K, and V in attention?

A: Q (Query), K (Key), and V (Value) are linear transformations of input embeddings used to calculate attention scores and outputs.

10. Q: What is the final output of a Transformer layer?

A: It is a transformed sequence of embeddings after attention and feed-forward processing, used for prediction or passed to the next layer.